

```

1s save_model(tuned_model, 'diabetes_prediction')

Transformation Pipeline and Model Successfully Saved
(Pipeline(memory=Memory(location=None),
  steps=[('numerical_imputer',
    TransformerWrapper(exclude=None,
      include=[
        'Number of times pregnant',
        'Plasma glucose concentration a 2 '
        'hours in an oral glucose '
        'tolerance test',
        'Diastolic blood pressure (mm Hg)',
        'Triceps skin fold thickness (mm)',
        '2-Hour serum insulin (mu U/ml)',
        'Body mass index (weight in '
        'kg/(height in m)^2)',
        'Diabetes pedigree...'
      ],
      RandomForestClassifier(bootstrap=True, ccp_alpha=0.0,
        class_weight=None, criterion='gini',
        max_depth=None, max_features='sqrt',
        max_leaf_nodes=None, max_samples=None,
        min_impurity_decrease=0.0,
        min_samples_leaf=1, min_samples_split=2,
        min_weight_fraction_leaf=0.0,
        monotonic_cst=None, n_estimators=100,
        n_jobs=-1, oob_score=False,
        random_state=123, verbose=0,
        warm_start=False))),
    verbose=False),
  'diabetes_prediction.pkl')

Start coding or generate with AI.

```

Figure 9: Saving the model

```

'2-Hour serum insulin (mu U/ml)':
[20],
'Body mass index (weight in kg/
(height in m)^2)': [34.3],
'Diabetes pedigree function':
[0.777],
'Age (years)': [45]
})

from pycaret.classification import
predict_model

# Predict with the model
predictions = predict_model(loaded_
model, data=new_data)
print(predictions)

```

As we can see in the figure, the predicted class says 0.66. This means that for the values we gave, the probability that the person may turn diabetic is more than 50%.

Building an ML model using PyCaret is as simple as this. You can explore other datasets, algorithms and use cases to gain further experience and learn more. 

```

0s [15] loaded_model = load_model('diabetes_prediction')

Transformation Pipeline and Model Successfully Loaded

```

Figure 10: Loading back the model

```

0s from pycaret.classification import predict_model

# Predict with the model
predictions = predict_model(loaded_model, data=new_data)
print(predictions)

Transformation Pipeline and Model Successfully Loaded

Number of times pregnant \
0 4

Plasma glucose concentration a 2 hours in an oral glucose tolerance test \
0 76

Diastolic blood pressure (mm Hg) Triceps skin fold thickness (mm) \
0 35 0

2-Hour serum insulin (mu U/ml) \
0 20

Body mass index (weight in kg/(height in m)^2) Diabetes pedigree function \
0 34.299999 0.777

Age (years) prediction_label prediction_score
0 45 0 0.66

```

Figure 11: Making a prediction using sample data

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The author is a data scientist, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.